Unit 5 - Ordinary Differential Equations - 1st Sem, B.Tech. 2023

Department of Mathematics, School of Engineering, Dayananda Sagar University

Contents

1 Introduction **1**
1.1 Second-order linear ODEs . . . 2
1.2 General Solutions to second-order linear ODEs . . . 2

2 Homogeneous Second-Order Linear ODEs with Constant Coefficients **3**
2.1 Solution overview . . . 3
2.2 Examples . . . 4
2.3 Initial Value Problems (IVPs) . . . 5
2.4 Exercises . . . 6

3 Non-homogeneous ODEs - Method of Undetermined Coefficients **6**
3.1 Solution overview : Particular solution/integral . . . 6
3.2 Examples . . . 7
3.3 Exercises . . . 9

4 Non-homogeneous ODEs - Variation of parameters **9**
4.1 Solution overview: Variation of Parameters . . . 10
4.2 Examples . . . 10
4.3 Exercises . . . 11

5 Hints and Solutions **12**

References **20**

1 Introduction

The notes here are a concise introduction to solving second-order linear ordinary differential equations and comprise the last module of the course titled "Linear algebra and Differential equations". The previous four modules, [Shw23], [Rak23], [Rac23] and [Sri23] cover a short introduction to Linear algebra.

Differential equations are mathematical equations that describe relationships between functions and their

derivatives. They're used to model dynamic systems where the rate of change of a quantity is related to its current state. These equations are crucial in understanding innumerable phenomena, from population growth and heat diffusion to electrical circuits and quantum mechanics. In the realm of differential equations, there are two main categories: **Ordinary differential equations** (ODEs) and **Partial differential equations** (PDEs).

ODEs involve functions of a single variable and their derivatives, whereas PDEs involve functions of multiple variables and their partial derivatives. As part of this domain, we have linear ordinary differential equations, where the dependent variable and its derivatives appear linearly. Specifically, second-order linear ODEs are of significant interest as they involve the second derivative of the dependent variable, often encountered in physical systems exhibiting inertia, oscillations, or damped behavior.

The main prerequisites for these notes include Differential and Integral calculus. Also, an exposure to first-order ordinary differential equations would be helpful. The usual references for an (engineering mathematics) ODE course are [Kre20], [Gre12] but a delightful (pure mathematical) one worth looking at is [CA92].

1.1 Second-order linear ODEs

A second-order linear ODE takes the following form wherein y represents the unknown function of x, while $p(x)$, $q(x)$, and $r(x)$ are given functions of x.

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = r(x)$$

.

We represent the above equation more succinctly as follows.

$$y'' + p(x)y' + q(x)y = r(x) \tag{1.1.1}$$

In these notes, however, **we shall only deal with ODEs with constant coefficients**, i.e., $p(x) = a$, $q(x) = b$ in 1.1.

1.2 General Solutions to second-order linear ODEs

The general approach to finding solutions for a second-order linear ordinary differential equation of the form:

$$y'' + ay' + by = r(x) \tag{1.2.1}$$

involves these steps:

1. **Homogeneous Solution:** First, consider the associated homogeneous equation, where $r(x) = 0$:

   $$y'' + ay' + by = 0$$

   Solve this equation to find the **complementary function** y_c by using methods like assuming solutions of the form $y = e^{rx}$ and finding the roots of the characteristic equation.

2. **Particular Solution:** Then, focus on finding a particular solution y_p that satisfies the non-homogeneous equation $y'' + ay' + by = r(x)$. The form of y_p depends on the specific form of $r(x)$ and may involve methods such as variation of parameters or undetermined coefficients.

3. **General Solution:** Once you have the **complementary function** y_c and the **particular solution** y_p, the general solution y_{general} to the original non-homogeneous equation is:

$$y_{\text{general}} = y_c + y_p$$

The exact method to find y_c and y_p can vary based on the forms of $p(x)$, $q(x)$, and $r(x)$. Solving for the complementary function typically involves techniques for solving homogeneous linear differential equations, while solving for the particular solution involves methods specific to the non-homogeneous term.

2 Homogeneous Second-Order Linear ODEs with Constant Coefficients

In this section, we explore solving homogeneous second-order linear ordinary differential equations (ODEs) with constant coefficients of the form:

$$y'' + ay' + by = 0 \tag{2.0.1}$$

where a and b are constants.

2.1 Solution overview

To solve these equations, we assume a solution in the form $y = e^{\lambda x}$. Substituting this into the ODE gives us the equation below.

$$\lambda^2 e^{\lambda x} + a\lambda e^{\lambda x} + be^{\lambda x} = 0 \tag{2.1.1}$$

Here, $e^{\lambda x}$ is always positive, allowing us to express the equation solely in terms of λ, which is what we refer to as the **characteristic equation**.

$$\lambda^2 + a\lambda + b = 0 \tag{2.1.2}$$

Solving this quadratic equation for λ yields the roots λ_1, λ_2.

$$\lambda_i = \frac{-a \pm \sqrt{a^2 - 4b}}{2} \tag{2.1.3}$$

Based on the nature of the roots, i.e., $a^2 - 4b\,(> \text{ or } = \text{ or } <)\,0$, the general solution can be expressed as follows wherein A and B are arbitrary scalars.

1. **Distinct Real Roots ($\lambda_1 \neq \lambda_2$):**
$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

2. **Repeated Real Roots ($\lambda_1 = \lambda_2$):**
$$y = (A + Bx)e^{\lambda x}$$

3. **Complex Roots ($\lambda = \alpha \pm \beta i$, $\beta > 0$):**
$$y = e^{\alpha x}(A\cos(\beta x) + B\sin(\beta x))$$

2.2 Examples

2.2.1 Distinct real roots

Consider the equation $y'' - 3y' + 2y = 0$.

The characteristic equation is $\lambda^2 - 3\lambda + 2 = 0$, which factors into $(\lambda - 1)(\lambda - 2) = 0$.

The roots are $\lambda_1 = 1$ and $\lambda_2 = 2$, yielding the general solution:
$$y = Ae^x + Be^{2x}$$

2.2.2 Repeated real roots

Solve the ODE $y'' - 4y' + 4y = 0$.

The characteristic equation is $\lambda^2 - 4\lambda + 4 = 0$, which factors to $(\lambda - 2)^2 = 0$.

This gives a repeated real root $\lambda = 2$, resulting in the general solution:
$$y = (A + Bx)e^{2x}$$

2.2.3 Complex roots

Solve the ODE $y'' + 2y' + 2y = 0$.

The characteristic equation is $\lambda^2 + 2\lambda + 2 = 0$, which has complex roots.

Solving for the roots yields $\lambda = -1 \pm i$, resulting in the general solution:
$$y = e^{-x}(A\cos(x) + B\sin(x))$$

2.2.4 Complex roots

Consider the equation $y'' + 4y = 0$.

The characteristic equation is $\lambda^2 + 4 = 0$, which factors into $(\lambda + 2i)(\lambda - 2i) = 0$.

The roots are $\lambda = -2i$ and $\lambda = 2i$, yielding the general solution:
$$y = (A\cos(2x) + B\sin(2x))$$

2.3 Initial Value Problems (IVPs)

When dealing with second-order linear ODEs, Initial Value Problems involve determining the specific solution that satisfies not only the differential equation but also specific initial conditions.

Consider the general form of the second-order linear ODE:

$$y'' + ay' + by = 0 \tag{2.3.1}$$

To solve an Initial Value Problem, additional information is provided, typically in the form of initial conditions for the function and its derivative at a specific point $x = x_0$ or $t = t_0$:

$$y(x_0) = y_0 \tag{2.3.2}$$
$$y'(x_0) = y_1 \tag{2.3.3}$$

These initial conditions serve as constraints that help determine the particular solution that satisfies both the ODE and the given conditions. The process involves using the general solution obtained earlier and applying the initial values to solve for the arbitrary constants.

2.3.1 Example (IVP)

For instance, considering the equation $y'' - 3y' + 2y = 0$ with initial conditions $y(0) = 1$ and $y'(0) = 0$ we get the general solution:

$$y = Ae^x + Be^{2x}$$

Using the initial conditions, let's first write out the functions $y(x)$ and $y'(x)$ using the general solution $y = Ae^x + Be^{2x}$ obtained earlier:

$$y(x) = Ae^x + Be^{2x} \tag{2.3.4}$$
$$y'(x) = Ae^x + 2Be^{2x} \tag{2.3.5}$$

Now, substitute $x = 0$ into these expressions to find A and B.

$$y(0) = A + B = 1 \tag{2.3.6}$$
$$y'(0) = A + 2B = 0 \tag{2.3.7}$$

Solving these simultaneous equations allows for the determination of the constants A and B specific to this Initial Value Problem.

We have $A = 2$ and $B = -1$ which gives us the final solution below.

$$y = 2e^x - e^{2x}$$

2.4 Exercises

Solve the following ODEs.

2.4.1 $10y'' - 7y' + 1.2y = 0$

2.4.2 $y'' + 4\pi y' + 4\pi^2 y = 0$

2.4.3 $y'' + 2y' + 5y = 0$

2.4.4 $y'' + 2.4y' + 4y = 0$

Solve the following IVPs.

2.4.5 $y'' + 11y' + 24y = 0 \qquad y(0) = 0 \qquad y'(0) = -7$

2.4.6 $y'' + 3y' - 10y = 0 \qquad y(0) = 4 \qquad y'(0) = -2$

2.4.7 $3y'' + 2y' - 8y = 0 \qquad y(0) = -6 \qquad y'(0) = -18$

2.4.8 $4y'' - 5y' = 0 \qquad y(-2) = 0 \qquad y'(-2) = 7$

3 Non-homogeneous ODEs - Method of Undetermined Coefficients

Consider a linear non-homogeneous ordinary differential equation (ODE) with constant coefficients:

$$y'' + ay' + by = r(x) \tag{3.0.1}$$

where a and b are constants, and $r(x)$ is a function of x.

The general solution to this requires the **complementary function** y_c (which is a solution to the corresponding homogeneous ODE) and the **particular solution** y_p.

$$y_{\text{general}} = y_c + y_p$$

3.1 Solution overview : Particular solution/integral

We choose a form for y_p similar to $r(x)$ (using the table below), but with unknown coefficients to be determined by substituting that y_p and its derivatives into the ODE.

1. **Basic Rule:** If $r(x)$ in 3.0.1 is one of the functions in the first column in table 1, choose y_p in the same line and determine its undetermined coefficients by substituting y_p and its derivatives into 3.0.1.

2. **Modification Rule:** If a term in your choice for y_p happens to be a solution of the homogeneous ODE corresponding to 3.0.1, multiply this term by x (or by x^2 if this solution corresponds to a double root of the characteristic equation of the homogeneous ODE).

Table 1: Particular Solutions for Different Non-Homogeneous Terms

$r(x)$ **(Non-Homogeneous Term)**	$y_p(x)$ **(Particular Solution)**
$ke^{\gamma x}$	$Ce^{\gamma x}$
kx^n $(n \geq 0, 1, \ldots)$	$K_n x^n + K_{n-1}x^{n-1} + \ldots + K_1 x + K_0$
$k\cos(\omega x)$	$K\cos(\omega x) + M\sin(\omega x)$
$k\sin(\omega x)$	$K\cos(\omega x) + M\sin(\omega x)$
$ke^{\alpha x}\cos(\omega x)$	$e^{\alpha x}(K\cos(\omega x) + M\sin(\omega x))$
$ke^{\alpha x}\sin(\omega x)$	$e^{\alpha x}(K\cos(\omega x) + M\sin(\omega x))$

3. **Sum Rule:** If $r(x)$ is a sum of functions in the first column of table 1, choose for y_p the sum of the functions in the corresponding lines of the second column.

3.2 Examples

(3.2.1) Solve $y'' + y = 0.001x^2$.

The general solution of the homogeneous ODE is:

$$y_h = A\cos x + B\sin x$$

For the non-homogeneous ODE, choosing $y_p = K_2x^2 + K_1x + K_0$, we get:

$$y_p'' + y_p = 2K_2 + K_2x^2 + K_1x + K_0 = 0.001x^2$$

Equating coefficients, we find:

$$K_2 = 0.001, \quad K_1 = 0, \quad 2K_2 + K_0 = 0$$

Hence, $K_0 = -2 \times 0.001 = -0.002$

This gives $y_p = 0.001x^2 - 0.002$

Therefore, the general solution of the differential equation $y'' + y = 0.001x^2$ is:

$$y = A\cos x + B\sin x + 0.001x^2 - 0.002$$

(3.2.2) Solve $y'' + 3y' + 2.25y = -10e^{-1.5x}$

Step 1: General solution of the homogeneous ODE: The characteristic equation of the homogeneous ODE is $\lambda^2 + 3\lambda + 2.25 = (\lambda + 1.5)^2 = 0$. Hence, the homogeneous ODE has the general solution:

$$y_h = (C_1 + C_2x)e^{-1.5x}$$

Step 2: Solution y_p of the non-homogeneous ODE: The function $e^{-1.5x}$ on the right corresponds to a solution of the homogeneous ODE, representing a double root of the characteristic equation. According to the Modification Rule in 3.1, we multiply our choice function by x^2 and choose:

$$y_p = Cx^2e^{-1.5x}$$

The derivatives of y_p are:

$$y_p' = 2Cxe^{-1.5x} - 1.5Cx^2e^{-1.5x}$$
$$y_p'' = 2Ce^{-1.5x} - 6Cxe^{-1.5x} - 3Cx^2e^{-1.5x}$$

Substituting these derivatives into the differential equation $y'' + 3y' + 2.25y = -10e^{-1.5x}$ we get:

$$2Ce^{-1.5x} - 6Cxe^{-1.5x} - 3Cx^2e^{-1.5x} + 3(2Cxe^{-1.5x} - 1.5Cx^2e^{-1.5x}) + 2.25x^2(Ce^{-1.5x}) = -10e^{-1.5x}$$

We cancel out $e^{-1.5x}$ to get:

$$2C - 6Cx - 3Cx^2 + 3(2Cx - 1.5Cx^2) + 2.25x^2C = -10$$

Solving this equation for C by comparing co-efficients yields $C = -5$.

Therefore, the particular solution is $y_p = -5x^2e^{-1.5x}$. Hence, the given ODE has the general solution:

$$y = y_h + y_p = (C_1 + C_2x)e^{-1.5x} - 5x^2e^{-1.5x}$$

(3.2.3) Solve $y'' + 2y' + 5y = e^{0.5x} + 40\cos 10x - 190\sin 10x$.

Step 1: Complementary function: The characteristic equation for the homogeneous equation:

$$\lambda^2 + 2\lambda + 5 = 0$$

Solving this quadratic equation gives complex roots:

$$\lambda = -1 + 2i \text{ and } -1 - 2i$$

Therefore, the complementary function is:

$$y_c = e^{-x}(C_1\cos(2x) + C_2\sin(2x))$$

Step 2: Particular solution using undetermined coefficients: Assume the particular solution as:

$$y_p = Ae^{0.5x} + B\cos(10x) + C\sin(10x)$$

Differentiating y_p twice:

$$y_p' = \frac{A}{2}e^{0.5x} - 10B\sin(10x) + 10C\cos(10x)$$

$$y_p'' = \frac{A}{4}e^{0.5x} - 100B\cos(10x) - 100C\sin(10x)$$

Substitute these derivatives into the differential equation:

$\frac{A}{4}e^{0.5x} - 100B\cos(10x) - 100C\sin(10x) + 2\left(\frac{A}{2}e^{0.5x} - 10B\sin(10x) + 10C\cos(10x)\right) +$

$5\left(Ae^{0.5x} + B\cos(10x) + C\sin(10x)\right) = e^{0.5x} + 40\cos(10x) - 190\sin(10x)$

Equate coefficients of like terms to solve for A, B, and C (we use linear independence of $\{e^x, \sin(x), \cos(x)\}$ here).

We obtain $A = 0.16, B = 0$ and $C = 2$ so that the final general solution is:

$$y = y_c + y_p$$

$$y = e^{-x}(C_1\cos(2x) + C_2\sin(2x)) + 0.16e^{0.5x} + 2\sin(10x)$$

3.3 Exercises

Solve the following using the method of undetermined coefficients.

3.3.1 $y'' - y' - 6y = e^{-x} - 7\cos(x)$

3.3.2 $y'' - 3y' = e^{3x} - 12x$

3.3.3 $y'' - y = e^x + x^2$

3.3.4 $y'' - y' = 5e^x - \sin(2x)$

4 Non-homogeneous ODEs - Variation of parameters

In this section, we focus on solving non-homogeneous linear second-order ordinary differential equations (ODEs) with constant coefficients of the form:

$$y'' + ay' + by = r(x) \tag{4.0.1}$$

where a and b are constants and $r(x)$ represents a non-zero function.

4.1 Solution overview: Variation of Parameters

For non-homogeneous equations, the method of variation of parameters is often employed. Given the complementary function y_c obtained from the corresponding homogeneous equation (as discussed in Section 2), the particular solution y_p is expressed as a linear combination of the complementary function multiplied by unknown functions of x:

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x) \tag{4.1.1}$$

Here, y_1 and y_2 are (linearly independent) solutions of the homogeneous equation, and $u_1(x)$ and $u_2(x)$ are functions to be determined.

To find $u_1(x)$ and $u_2(x)$, the Wronskian $W(x)$ is employed:

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1(x)y_2'(x) - y_2(x)y_1'(x) \tag{4.1.2}$$

The particular solution y_p is then obtained using the following formulae wherein the constants that come from the integrals can be assumed to be zero without loss of generality.

$$u_1(x) = -\int \frac{y_2(x)r(x)}{W(x)}\,dx, \quad u_2(x) = \int \frac{y_1(x)r(x)}{W(x)}\,dx \tag{4.1.3}$$

Finally, the general solution to the non-homogeneous equation is:

$$y(x) = y_c(x) + y_p(x) \tag{4.1.4}$$

4.2 Examples

(4.2.1) Consider the equation $y'' + y = \sec(x)$.

The associated homogeneous equation is $y'' + y = 0$ with the solution $y_c = A\cos(x) + B\sin(x)$.

For the particular solution, let $y_1 = \cos(x)$ and $y_2 = \sin(x)$.

The Wronskian $W(x)$ is $W(x) = \cos(x) \cdot \cos(x) - \sin(x) \cdot (-\sin(x)) = 1$.

Using the formulas for $u_1(x)$ and $u_2(x)$:

$$u_1(x) = -\int \frac{\sin(x) \cdot \sec(x)}{1}\,dx = -\int \tan(x)\,dx = \ln|\cos(x)| + C_1$$

$$u_2(x) = \int \frac{\cos(x) \cdot \sec(x)}{1}\,dx = \int dx = x + C_2$$

Therefore, a particular solution $y_p(x) = \ln|\cos(x)|\cos(x) + x\sin(x)$.

The general solution is $y(x) = A\cos(x) + B\sin(x) + \cos(x)\ln|\cos(x)| + x\sin(x)$.

(4.2.2) Find a general solution to the following differential equation.

$$y'' - 2y' + y = \frac{e^t}{t^2+1}$$

We first need the complementary solution for this differential equation. The complementary solution is,

$$y_c(t) = c_1 e^t + c_2 t e^t$$

So, we have

$$y_1(t) = e^t \quad y_2(t) = te^t$$

The Wronskian of these two functions is

$$W = \begin{vmatrix} e^t & te^t \\ e^t & (1+t)e^t \end{vmatrix} = e^{2t}$$

$$\begin{aligned} y_p(t) &= -\mathbf{e}^t \int \frac{t\mathbf{e}^t\,\mathbf{e}^t}{\mathbf{e}^{2t}\,(t^2+1)}\,dt + t\mathbf{e}^t \int \frac{\mathbf{e}^t\,\mathbf{e}^t}{\mathbf{e}^{2t}\,(t^2+1)}\,dt \\ &= -\mathbf{e}^t \int \frac{t}{t^2+1}\,dt + t\mathbf{e}^t \int \frac{1}{t^2+1}\,dt \\ &= -\frac{1}{2}\mathbf{e}^t \ln\left(1+t^2\right) + t\mathbf{e}^t \tan^{-1}(t) \end{aligned}$$

The general solution is,

$$y(t) = c_1\mathbf{e}^t + c_2 t\mathbf{e}^t - \frac{1}{2}\mathbf{e}^t \ln\left(1+t^2\right) + t\mathbf{e}^t \tan^{-1}(t)$$

4.3 Exercises

4.3.1 $y'' + y = \cot x$

4.3.2 $y'' + y = \sec x \tan x$

4.3.3 $y'' + y = \mathrm{cosec} x$

4.3.4 $y'' + y = \tan x$

5 Hints and Solutions

2.4.1 $10y'' - 7y' + 1.2y = 0$

The standard form of the differential equation is:

$$y'' - 0.7y' + 0.12y = 0$$

The characteristic equation is given by:

$$\lambda^2 - 0.7\lambda + 0.12 = 0$$

Solving the characteristic equation yields roots $\lambda_1 = 0.4$ and $\lambda_2 = 0.3$. Hence, the corresponding general solution is:

$$y(x) = c_1 e^{0.4x} + c_2 e^{0.3x}$$

2.4.2 $y'' + 4\pi y' + 4\pi^2 y = 0$

The characteristic equation is given by:

$$\lambda^2 + 4\pi\lambda + 4\pi^2 = 0$$

Solving the characteristic equation yields a double root at $\lambda = -2\pi$. Therefore, the corresponding general solution is:

$$y(x) = (c_1 + c_2 x)e^{-2\pi x}$$

2.4.3 $y'' + 2y' + 5y = 0$

The characteristic equation is given by:

$$\lambda^2 + 2\lambda + 5 = 0$$

Solving the characteristic equation yields complex roots, which can be expressed as:

$$\lambda = -1 + 2i \quad \text{and} \quad \lambda = -1 - 2i$$

Therefore, the corresponding general solution is:

$$y(x) = e^{-x}(c_1 \cos(2x) + c_2 \sin(2x))$$

2.4.4 $y'' + 2.4y' + 4y = 0$

The characteristic equation is given by:

$$\lambda^2 + 2.4\lambda + 4 = 0$$

Solving the characteristic equation yields real and distinct roots, which can be expressed as:

$$\lambda = -1.2 + 1.6i \quad \text{and} \quad \lambda = -1.2 - 1.6i$$

Therefore, the corresponding general solution is:

$$y(x) = e^{-1.2x}(c_1 \cos(1.6x) + c_2 \sin(1.6x))$$

2.4.5 $y'' + 11y' + 24y = 0 \qquad y(0) = 0 \qquad y'(0) = -7$

The characteristic equation is:

$$r^2 + 11r + 24 = 0$$
$$(r+8)(r+3) = 0$$

Its roots are $r_1 = -8$ and $r_2 = -3$, so the general solution and its derivative are:

$$y(t) = c_1 e^{-8t} + c_2 e^{-3t}$$
$$y'(t) = -8c_1 e^{-8t} - 3c_2 e^{-3t}$$

Now, plug in the initial conditions to get the following system of equations.

$$0 = y(0) = c_1 + c_2$$
$$-7 = y'(0) = -8c_1 - 3c_2$$

Solving this system gives $c_1 = \frac{7}{5}$ and $c_2 = -\frac{7}{5}$. The actual solution to the differential equation is then:

$$y(t) = \frac{7}{5}e^{-8t} - \frac{7}{5}e^{-3t}$$

2.4.6 $y'' + 3y' - 10y = 0 \qquad y(0) = 4 \qquad y'(0) = -2$

The characteristic equation is:

$$r^2 + 3r - 10 = 0$$
$$(r+5)(r-2) = 0$$

Its roots are $r_1 = -5$ and $r_2 = 2$, so the general solution and its derivative are:

$$y(t) = c_1 e^{-5t} + c_2 e^{2t}$$
$$y'(t) = -5c_1 e^{-5t} + 2c_2 e^{2t}$$

Now, plug in the initial conditions to get the following system of equations.

$$4 = y(0) = c_1 + c_2$$
$$-2 = y'(0) = -5c_1 + 2c_2$$

Solving this system gives $c_1 = \frac{10}{7}$ and $c_2 = \frac{18}{7}$. The actual solution to the differential equation is then:

$$y(t) = \frac{10}{7} e^{-5t} + \frac{18}{7} e^{2t}$$

2.4.7 $3y'' + 2y' - 8y = 0 \qquad y(0) = -6 \qquad y'(0) = -18$

The characteristic equation is:

$$3r^2 + 2r - 8 = 0$$
$$(3r - 4)(r + 2) = 0$$

Its roots are $r_1 = \frac{4}{3}$ and $r_2 = -2$ and so the general solution and its derivative is:

$$y(t) = c_1 \mathbf{e}^{\frac{4t}{3}} + c_2 \mathbf{e}^{-2t}$$
$$y'(t) = \frac{4}{3} c_1 \mathbf{e}^{\frac{4t}{3}} - 2c_2 \mathbf{e}^{-2t}$$

Now, plug in the initial conditions to get the following system of equations.

$$-6 = y(0) = c_1 + c_2$$
$$-18 = y'(0) = \frac{4}{3} c_1 - 2c_2$$

Solving this system gives $c_1 = -9$ and $c_2 = 3$. The actual solution to the differential equation is then.

$y(t) = -9\mathbf{e}^{\frac{4t}{3}} + 3\mathbf{e}^{-2t}$

2.4.8 $4y'' - 5y' = 0 \qquad y(-2) = 0 \qquad y'(-2) = 7$

The characteristic equation is:

$$4r^2 - 5r = 0$$
$$r(4r - 5) = 0$$

The roots of this equation are $r_1 = 0$ and $r_2 = \frac{5}{4}$. Here is the general solution as well as its derivative.

$$y(t) = c_1 e^0 + c_2 e^{\frac{5t}{4}} = c_1 + c_2 e^{\frac{5t}{4}}$$
$$y'(t) = \frac{5}{4} c_2 e^{\frac{5t}{4}}$$

Up to this point, all of the initial conditions have been at $t = 0$, and this one isn't. Don't get too locked into initial conditions always being at $t = 0$ and you just automatically use that instead of the actual value for a given problem.

So, plugging in the initial conditions gives the following system of equations to solve.

$$0 = y(-2) = c_1 + c_2 e^{-\frac{5}{2}}$$
$$7 = y'(-2) = \frac{5}{4} c_2 e^{-\frac{5}{2}}$$

Solving this gives:

$c_1 = -\frac{28}{5}$ and $c_2 = \frac{28}{5} e^{\frac{5}{2}}$

The solution to the differential equation is then:

$y(t) = -\frac{28}{5} + \frac{28}{5} e^{\frac{5}{2}} e^{\frac{5t}{4}} = -\frac{28}{5} + \frac{28}{5} e^{\frac{5t}{4} + \frac{5}{2}}$

3.3.1 $y'' - y' - 6y = e^{-x} - 7\cos(x)$

The complementary function $y_c(x)$ is found from the characteristic equation, resulting in:

$$y_c(x) = C_1 e^{3x} + C_2 e^{-2x}$$

For the particular solution, we get from the sum rule:

$$y_p(x) = Ae^{-x} + B\sin(x) + C\cos(x)$$

Then,

$$y_p'(x) = -Ae^{-x} + B\cos(x) - C\sin(x)$$

$$y_p''(x) = Ae^{-x} - B\sin(x) - C\cos(x)$$

Substituting these into the differential equation gives:

$$-4Ae^{-x} - (7B + C)\sin(x) - (B - 7C)\cos(x) = e^{-x} - 7\cos(x)$$

Equating coefficients, we find:

$$A = \frac{-1}{4}, \quad B = \frac{7}{50}, \quad C = \frac{49}{50}$$

Therefore, the particular solution is:

$$y_p(x) = \frac{-1}{4}e^{-x} + \frac{7}{50}\sin(x) + \frac{49}{50}\cos(x)$$

The general solution is the sum of the complementary function and the particular solution:

$$y(x) = C_1e^{3x} + C_2e^{-2x} - \frac{1}{4}e^{-x} + \frac{7}{50}\sin(x) + \frac{49}{50}\cos(x)$$

3.3.2 $y'' - 3y' = e^{3x} - 12x$

Solving the auxiliary equation $\lambda^2 - 3\lambda = 0$ gives $\lambda = 0, \lambda = 3$. Hence, the complementary function is:

$$y_c(x) = c_1 + c_2e^{3x}$$

For the particular solution, we have (using the modification and the sum rule):

$$y_p(x) = Axe^{3x} + Bx^2 + Cx$$

(Note that the modification rule also applies to $Bx + C$ as C is a constant and c_1, which is also a constant, appeared as a solution of the homogeneous equation.)

Then,

$$y_p'(x) = Ae^{3x} + 3Axe^{3x} + 2Bx + C$$
$$y_p''(x) = 6Ae^{3x} + 9Axe^{3x} + 2B$$

Setting this into the differential equation gives:

$$6Ae^{3x} + 9Axe^{3x} + 2B - 3(Ae^{3x} + 3Axe^{3x} + 2Bx + C)$$
$$= 3Ae^{3x} - 6Bx + 2B - 3C = e^{3x} - 12x$$

From this, we find $A = \frac{1}{3}$, $B = 2$, and $C = \frac{4}{3}$. Therefore, the particular solution is:

$$y_p(x) = \frac{1}{3}xe^{3x} + 2x^2 + \frac{4}{3}x$$

The general solution is the sum of the complementary function and the particular solution:

$$y(x) = c_1 + c_2 e^{3x} + \frac{1}{3} x e^{3x} + 2x^2 + \frac{4}{3} x$$

3.3.3 $y'' - y = e^x + x^2$

The auxiliary equation is $\lambda^2 - 1 = 0$, resulting in $(\lambda + 1)(\lambda - 1) = 0$. This gives the complementary function:

$$y_c(x) = c_1 e^x + c_2 e^{-x}$$

For the particular solution, we have using modification rule for e^x and the sum rule:

$$y_p(x) = Axe^x + Bx^2 + Cx + D$$

Then,

$$y_p'(x) = Ae^x + Axe^x + 2Bx + C$$
$$y_p''(x) = 2Ae^x + Axe^x + 2B$$

Substituting these derivatives into the differential equation:

$$2Ae^x + Axe^x + 2B - (Axe^x + Bx^2 + Cx + D) = e^x + x^2$$

Equating coefficients:

$$A = \frac{1}{2}$$
$$B = 1$$
$$C = 0$$
$$2B - D = 0 \implies D = 2$$

The particular solution is thus:

$$y_p(x) = \frac{1}{2} x e^x + x^2 + 2$$

The general solution is:

$$y(x) = y_c(x) + y_p(x) = c_1 e^x + c_2 e^{-x} + \frac{1}{2} x e^x + x^2 + 2$$

3.3.4 $y'' - y' = 5e^x - \sin(2x)$

The auxiliary equation is $\lambda^2 - \lambda = 0$, resulting in $\lambda(\lambda - 1) = 0$. This gives the complementary function:

$$y_c(x) = c_1 e^x + c_2$$

For the particular solution, we take $y_p(x) = Axe^x + B\cos(2x) + C\sin(2x)$. Then,

$$y_p'(x) = Ae^x + Axe^x - 2B\sin(2x) + 2C\cos(2x)$$

$$y_p''(x) = 2Ae^x + Axe^x - 4B\cos(2x) - 4C\sin(2x)$$

Substituting these derivatives into the differential equation:

$$2Ae^x + Axe^x - 4B\cos(2x) - 4C\sin(2x) - (Ae^x + Axe^x - 2B\sin(2x) + 2C\cos(2x)) = 5e^x - \sin(2x)$$

Equating coefficients:

$$A = 5$$

$$2B - 4C = -1$$

$$-4B - 2C = 0$$

Therefore, $A = 5$, $B = -\frac{1}{10}$ and $C = \frac{1}{5}$. The particular solution is then:

$$y_p(x) = 5xe^x - \frac{1}{10}\cos(2x) + \frac{1}{5}\sin(2x)$$

Hence, the general solution is:

$$y(x) = c_1 e^x + c_2 + 5xe^x - \frac{1}{10}\cos(2x) + \frac{1}{5}\sin(2x)$$

4.3.1 $y'' + y = \cot x$

The corresponding homogeneous equation gives the following:

$$\begin{aligned}
&\lambda^2 + 1 = 0 \\
\implies &\lambda = \pm i \\
\implies &y_c = c_1 \cos x + c_2 \sin x \\
\implies &W = \cos x \cos x - (-\sin x)\sin x = 1
\end{aligned}$$

Thus we obtain,

$$\begin{aligned}
y_p(x) &= -\cos x \int \cos x \quad + \quad \sin x \int \cos x \cdot \cot x \\
&= -\cos x \sin x \quad + \quad \sin x \left(-\ln\left(|\operatorname{cosec} x + \cot x|\right) + \cos x\right) \\
&= \quad \sin x \ \ln\left(|\operatorname{cosec} x - \cot x|\right)
\end{aligned}$$

And the general solution $y_{\text{gen}} = y_c + y_p$.

4.3.2 $y'' + y = \sec x \tan x$

$$\begin{aligned}
&\lambda^2 + 1 = 0 \\
\Longrightarrow\ &\lambda = \pm i \\
\Longrightarrow\ &y_c = c_1 \cos x + c_2 \sin x \\
\Longrightarrow\ &W = \cos x \cos x - (-\sin x) \sin x = 1
\end{aligned}$$

Thus we obtain,

$$\begin{aligned}
y_p(x) &= -\cos x \int \tan^2 x \quad + \quad \sin x \int \tan x \\
&= -\cos x\,(\tan x - x) \quad + \quad \sin x\ \ln|\sec x|
\end{aligned}$$

And the general solution $y_{\text{gen}} = y_c + y_p$.

4.3.3 $y'' + y = \operatorname{cosec} x$

$$\begin{aligned}
&\lambda^2 + 1 = 0 \\
\Longrightarrow\ &\lambda = \pm i \\
\Longrightarrow\ &y_c = c_1 \cos x + c_2 \sin x \\
\Longrightarrow\ &W = \cos x \cos x - (-\sin x) \sin x = 1
\end{aligned}$$

Thus we obtain,

$$\begin{aligned}
y_p(x) &= -\cos x \int \sin x \operatorname{cosec} x \quad + \quad \sin x \int \cot x \\
&= -x\ \cos x \quad + \quad \sin x\ \ln|\sin x|
\end{aligned}$$

And the general solution $y_{\text{gen}} = y_c + y_p$.

4.3.4 $y'' + y = \tan x$

The corresponding homogeneous equation gives the following:

$$\begin{aligned}
&\lambda^2 + 1 = 0 \\
\Longrightarrow\ &\lambda = \pm i \\
\Longrightarrow\ &y_c = c_1 \cos x + c_2 \sin x \\
\Longrightarrow\ &W = \cos x \cos x - (-\sin x) \sin x = 1
\end{aligned}$$

Thus we obtain,

$$\begin{aligned}
y_p(x) &= -\cos x \int \sin x \, \tan x \quad + \quad \sin x \int \cos x \, \tan x \\
&= -\cos x \int (\sec x - \cos x) \quad - \quad \sin x \, \cos x \\
&= \quad -\cos x \, (\ln|\sec x + \tan x| - \sin x) \quad - \quad \sin x \cos x \\
&= \quad \cos x \, \ln|\sec x - \tan x|
\end{aligned}$$

And the general solution $y_{\text{gen}} = y_c + y_p$.

References

[CA92] R. Cooke and V.I. Arnold, *Ordinary Differential Equations*, Springer Textbook, Springer Berlin Heidelberg, 1992.

[Gre12] M.D. Greenberg, *Ordinary Differential Equations*, Wiley, 2012.

[Kre20] E. Kreyszig, *Advanced Engineering Mathematics*, Wiley, 2020.

[Rac23] Rachaita Dutta, Dept. of Math , DSU, *Linear transformations*, 2023.

[Rak23] Rakesh Singh, Dept. of Math , DSU, *Vector Spaces*, 2023.

[Shw23] Shweta Naik, Dept. of Math , DSU, *Linear systems and Matrix operations*, 2023.

[Sri23] Srimanta Maji, Dept. of Math , DSU, *Eigenvalues and Diagonalization*, 2023.